

Dynamic Guideline: WHO-Based Anthrax - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO Anthrax Guidance

- **Guideline Title:** *Anthrax: implementation guidance for clinicians*
- **Publication Date:** 26 August 2025
- **Document Type:** WHO normative clinical implementation guidance
- **Target Population:** Frontline health-care workers managing suspected or confirmed human anthrax, or exposed persons during naturally occurring outbreaks, especially in rural, vulnerable, resource-limited, or emergency settings.
- **Scope and Objectives:** Practical clinical guidance for naturally occurring human anthrax, focusing on diagnosis, antibiotic treatment, post-exposure prophylaxis, and infection prevention and control. WHO states that the document is designed for rapid field use and includes a poster/job aid summarizing key recommendations for clinicians [WHO 2025 anthrax implementation guidance](#).
- **Important Scope Limitation:** The implementation guide focuses on naturally occurring anthrax. The WHO clinician PDF states that intentional release or bioterrorism-related exposure is addressed in other WHO guidance, and that it does not provide guidance on monoclonal antibodies such as raxibacumab or obiltoximab, or anthrax immune globulin [WHO clinician implementation guidance PDF](#).

Exactly 5 WHO Anthrax-Related Guidance Documents Synthesized

No.	WHO Source	Date	Role in This Dynamic Guideline
1	<i>Anthrax: implementation guidance for clinicians</i>	26 August 2025	Latest WHO anthrax clinical implementation guidance for diagnosis, antibiotics, PEP, and IPC in naturally occurring human anthrax WHO 2025 anthrax implementation guidance
2	<i>Anthrax in humans and animals</i> , 4th edition	1 January 2008	Main comprehensive WHO/FAO/OIE guideline covering humans, animals, diagnosis, treatment/prophylaxis, disinfection/decontamination, control, and surveillance WHO 2008 publication page
3	<i>Guidelines for the Surveillance and Control of Anthrax in Humans and Animals</i>	1 January 1998	Earlier WHO surveillance/control guideline; frames anthrax as primarily a zoonosis affecting herbivores, with humans infected directly or indirectly from animals WHO 1998 guideline page
4	WHO rapid risk assessment: human anthrax outbreak in	WHO outbreak risk assessment	Operational One Health outbreak management: coordination, surveillance, laboratory confirmation, livestock vaccination, carcass disposal, RCCE, and

No.	WHO Source	Date	Role in This Dynamic Guideline
	North Kivu, DRC and Uganda		cross-border collaboration WHO rapid risk assessment PDF
5	WHO AFRO Integrated Disease Surveillance and Response technical guidelines	WHO AFRO surveillance guidance	Includes human anthrax among priority diseases/conditions and emphasizes One Health surveillance and outbreak response WHO AFRO IDSR PDF

Clarification on the Requested “Meaningful Youth Engagement” Title

The most recent WHO publication matching the title pattern “**Meaningful youth engagement in a WHO guideline development process: case study of WHO [] guideline**” is “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**”, published 11 December 2025. It is not an anthrax guideline and the available evidence does **not** show that youth engagement was used in WHO anthrax guideline development [WHO case study page](#), [WHO webinar page](#). Therefore, it is noted here for compliance with the requested title identification but is **not counted** among the five anthrax-related WHO guidance documents above.

1.2 Key Recommendations

Clinical Recommendations With GRADE Assessment

GRADE note: WHO anthrax documents located in the evidence summaries do not provide formal GRADE tables. The following GRADE ratings are an applied evidence appraisal using standard GRADE concepts: randomized human evidence is scarce because anthrax is rare; much evidence comes from observational case series, systematic reviews of case reports, animal studies, in vitro data, outbreak experience, and expert consensus. Recommendations may still be strong when disease severity is high, treatment benefit is biologically compelling, and harms are acceptable.

Recommendation	GRADE Evidence Quality	Strength
Suspect anthrax in compatible illness plus exposure to sick/dead livestock, contaminated animal products, undercooked meat from infected animals, contaminated heroin injection, occupational exposure, or possible aerosol exposure.	Moderate	Strong
Collect diagnostic specimens before antibiotics if doing so does not delay treatment; use lesion fluid/swabs or eschar-edge material, blood, CSF, respiratory secretions, ascitic fluid, or other site-specific specimens as clinically indicated.	Moderate	Strong
Do not delay treatment while awaiting laboratory confirmation when anthrax is clinically suspected.	Moderate	Strong

Recommendation	GRADE Evidence Quality	Strength
Treat localized uncomplicated cutaneous anthrax with an active single antibiotic such as ciprofloxacin, doxycycline, penicillin, or amoxicillin when susceptibility and clinical context support use.	Low–Moderate	Strong
Manage systemic anthrax—including gastrointestinal, inhalation, injection-related disease, bacteremia, meningitis, or cutaneous disease with systemic illness/extensive edema—as a medical emergency requiring inpatient care, combination antimicrobial therapy, and aggressive supportive care.	Low–Moderate	Strong
For systemic anthrax, use combination therapy; WHO outbreak guidance gives examples including penicillin or ciprofloxacin plus clindamycin for inhalation anthrax or sepsis, penicillin or ciprofloxacin plus gentamicin for gastrointestinal anthrax, and penicillin or ciprofloxacin plus rifampicin or vancomycin for anthrax meningitis.	Low	Strong
For exposed asymptomatic persons, provide post-exposure prophylaxis according to public-health direction; external CDC guidance supports antibiotics plus vaccine where available/indicated, with antibiotics up to 60 days after inhalational exposure risk.	Moderate	Strong
Apply standard infection prevention and control precautions for suspected or confirmed anthrax; person-to-person transmission is not typical, but contaminated lesion drainage or body fluids require careful handling.	Low–Moderate	Strong
Notify public-health authorities immediately and integrate human, animal, and environmental health sectors for case investigation, source control, animal vaccination, safe carcass disposal, community engagement, and cross-border coordination where relevant.	Moderate	Strong
Use antitoxin therapy for severe systemic/inhalation anthrax where available and indicated, based on non-WHO guidance; note that WHO 2025 clinician guidance does not provide specific recommendations on raxibacumab, obiltoximab, or anthrax immune globulin.	Low	Conditional

1.3 Guideline Development Methodology

WHO 2025 Clinical Implementation Guidance

- WHO describes the 2025 guidance as practical, up-to-date clinical guidance for frontline workers in rural, vulnerable, resource-limited, and emergency settings [WHO 2025 anthrax implementation guidance](#).
- The WHO clinician PDF states that it was developed as a practical summary of the 2008 WHO guideline *Anthrax in humans and animals* [WHO clinician implementation guidance PDF](#).
- It focuses on naturally occurring anthrax and addresses diagnosis, antibiotic treatment, post-exposure prophylaxis, and infection prevention and control [WHO 2025 anthrax implementation guidance](#).

WHO 2008 Comprehensive Guideline

- The 2008 fourth edition *Anthrax in humans and animals* was produced by WHO with FAO and OIE/WOAH involvement and covers etiology, ecology, anthrax in animals and humans, bacteriology, treatment/prophylaxis, control, and surveillance [WHO 2008 publication page](#).
- WHO states that the 2008 edition systematically reviewed scientific literature and relevant publications up to the end of 2007, including information emerging after the anthrax letter events [WHO 2008 publication page](#).
- The NCBI-hosted development section reports that the guideline incorporated conclusions from an expert meeting in Nice, France, in 2003, sought contributions from scientists with specific expertise, and underwent two rounds of rigorous peer review [NCBI Bookshelf: Development of these guidelines](#).

Consensus Method

- WHO 2008: wide consultation, expert contributions, correction/approval by contributors, and two rounds of peer review [NCBI Bookshelf: Development of these guidelines](#).
- WHO 2025: the provided summaries do not describe a formal consensus voting method.

Conflict of Interest Management

- The provided WHO anthrax source summaries do not identify WHO-specific conflict-of-interest procedures for the 2008 or 2025 anthrax documents.
- By comparison, CDC's 2023 anthrax guideline did document written financial disclosures and review of potential conflicts for CDC staff and external consultants [CDC 2023 anthrax guidelines](#).

1.4 WHO-Based Clinical Management Summary

Disease Definition and Transmission

Anthrax is a serious zoonotic bacterial infection caused by *Bacillus anthracis*. WHO describes anthrax as primarily a disease of herbivores, with humans almost invariably infected directly or indirectly from animals or animal products [WHO 1998 guideline page](#). WHO states that natural infection can occur through contact with infected animals or animal products, ingestion, inhalation, or entry of spores through skin breaks [WHO Europe anthrax Q&A](#).

Clinical Forms

WHO recognizes cutaneous, gastrointestinal, inhalation/pulmonary, injection-related, and systemic anthrax. The WHO clinician guidance defines systemic anthrax as any of: cutaneous anthrax with systemic illness, sepsis and/or extensive edema; gastrointestinal anthrax; inhalation anthrax; injection-related anthrax; or *B. anthracis* bacteremia [WHO clinician implementation guidance PDF](#).

The usual incubation period is **2–7 days**, although longer incubation can occur, especially after inhalational exposure [WHO clinician implementation guidance PDF](#).

Diagnosis

WHO clinician guidance lists diagnostic specimens including vesicle or blister fluid, ulcer or eschar specimens taken below the edges, blood, and site-specific specimens such as ascitic fluid or cerebrospinal fluid [WHO clinician implementation guidance PDF](#).

CDC adds that anthrax diagnosis can be confirmed by measuring lethal factor toxin in blood or by testing for *B. anthracis* in blood, skin lesion swab, CSF, or respiratory secretions, and advises collecting samples before antibiotics if this does not delay treatment [CDC Clinical Overview](#).

Treatment

- **Localized uncomplicated cutaneous anthrax:** WHO outbreak guidance states that localized cutaneous anthrax may be treated with a single agent such as ciprofloxacin, doxycycline, penicillin, or amoxicillin [WHO rapid risk assessment PDF](#).
- **Systemic anthrax:** WHO outbreak guidance supports combination therapy and supportive care. Examples include:
 - Penicillin or ciprofloxacin plus clindamycin for inhalation anthrax or sepsis.
 - Penicillin or ciprofloxacin plus gentamicin for gastrointestinal anthrax.
 - Penicillin or ciprofloxacin plus rifampicin or vancomycin for anthrax meningitis [WHO rapid risk assessment PDF](#).
- **Antitoxin:** WHO 2025 guidance does not cover raxibacumab, obiltoxaximab, or anthrax immune globulin. CDC- and Mayo Clinic–summarized practice supports antitoxin with antibiotics for severe toxin-mediated disease, especially inhalation/systemic anthrax [Mayo Clinic](#), [CDC 2023 MMWR](#).

Post-Exposure Prophylaxis

WHO 2025 guidance includes post-exposure prophylaxis as a core topic [WHO 2025 anthrax implementation guidance](#). CDC states that exposed persons require post-exposure prophylaxis with vaccines and antibiotics, and that antibiotics may be needed for up to 60 days because spores can remain dormant for prolonged periods [CDC Clinical Overview](#).

Infection Prevention and Control

WHO materials emphasize standard precautions. WHO IHR Annex 2 guidance notes that anthrax is not very contagious among humans, with potential contagiousness mainly through spore- or organism-contaminated body fluids [WHO IHR Annex 2 guidance PDF](#).

Surveillance and One Health Control

WHO 2008 states that anthrax surveillance aims to prevent or reduce livestock losses and prevent human disease. Cardinal surveillance/control actions include education, correct diagnosis, implementation of control measures, and reporting, supported by case definitions and laboratory backup [WHO IRIS anthrax PDF](#).

WHO outbreak guidance emphasizes One Health coordination, active case finding, laboratory confirmation, livestock vaccination in high-risk areas, safe carcass disposal, risk communication and community engagement, and cross-border collaboration [WHO rapid risk assessment PDF](#).

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Time Window

- **WHO latest anthrax guidance release:** 26 August 2025.
- **Evidence cutoff:** 29 April 2026.
- **Evidence interval assessed:** 27 August 2025 to 29 April 2026.

Search Strategy

Evidence sources prioritized:

- WHO official guidance and outbreak documents.
- CDC clinical guidance and 2023 systematic-review-based anthrax guideline.
- Clinical reference updates from Mayo Clinic and MSD/Merck Manual.
- Systematic reviews, clinical trials, and registries where available.

Databases and Sources Represented in the Provided Evidence Summaries

- WHO publications and WHO IRIS.
- CDC MMWR and CDC clinical pages.
- ClinicalTrials.gov.
- Mayo Clinic.
- MSD/Merck Manual Professional Edition.
- NCBI Bookshelf/StatPearls.
- WOAHA technical manual.
- Guideline Central.

Inclusion Criteria

- Anthrax diagnosis, treatment, prophylaxis, antitoxin, vaccine, infection prevention/control, surveillance, or outbreak management.
- Human clinical relevance prioritized.
- WHO sources prioritized; CDC and other authoritative sources used to supplement post-2025 evidence and areas not detailed in WHO guidance.

Exclusion Criteria

- Non-anthrax-specific evidence.
- Non-clinical opinion without actionable relevance.
- Evidence superseded by WHO 2025 or CDC 2023 unless needed for historical context.

Overall Finding

No new anthrax randomized trial, systematic review, or major guideline published between 27 August 2025 and 29 April 2026 was identified in the provided evidence summaries. The most current post-release clinical updates located were Mayo Clinic's anthrax diagnosis/treatment page updated 24 December 2025 and MSD/Merck Manual Professional Edition material carrying 2026 copyright, both consistent with existing CDC/WHO principles [Mayo Clinic](#), [MSD Manual Professional Edition](#).

2.2 Key New Studies and Evidence Sources

Study / Source	Design	Key Findings	GRADE Quality
Mayo Clinic anthrax diagnosis and treatment page, updated 24 December 2025	Expert clinical reference summarizing accepted practice	Diagnosis may include imaging, lumbar puncture, skin biopsy, and other tests; treatment includes antibiotics and, in some cases, antitoxin; advanced inhalation anthrax may not respond to antibiotics alone because toxin burden may be high Mayo Clinic .	Low
MSD/Merck Manual Professional Edition, 2026 copyright	Expert clinical reference	Supports single-agent therapy for uncomplicated cutaneous anthrax, prolonged prophylaxis after inhalational exposure, and antitoxin/immune globulin options for severe disease MSD Manual Professional Edition .	Low
CDC 2023 Guidelines for the Prevention and Treatment of Anthrax	Guideline based on systematic reviews and data analyses	Updated antimicrobial and antitoxin recommendations for PEP and treatment; evidence base included systematic reviews of in vitro antimicrobial activity, animal efficacy, antitoxin efficacy, and human survival CDC 2023 MMWR .	Moderate
CDC systematic review of hospitalized/fatal human anthrax cases within 2023 guideline	Systematic review of case reports, case series, and line lists	Identified 13,082 sources; 368 articles yielded 965 confirmed hospitalized or fatal anthrax patients for clinical and treatment analyses CDC 2023 MMWR .	Low–Moderate
ClinicalTrials.gov NCT01491607 BioThrax trial	Phase 3 vaccine trial, not newly published in 2025–2026	Evaluated immunogenicity and safety of BioThrax anthrax vaccine in healthy adults after three doses; relevant to PEP vaccine evidence but not new in the post-WHO-2025 window ClinicalTrials.gov .	Moderate
CDC Clinical Overview of Anthrax	Public-health clinical guidance	Confirms diagnosis by lethal factor toxin testing or <i>B. anthracis</i> testing from relevant specimens; recommends sample collection before antibiotics if no treatment delay; exposed patients require PEP with vaccines and antibiotics CDC Clinical Overview .	Moderate

Study / Source	Design	Key Findings	GRADE Quality
WHO rapid risk assessment for DRC/Uganda outbreak	Outbreak operational evidence/guidance	Reinforces One Health response, active case finding, laboratory confirmation, livestock vaccination, carcass disposal, RCCE, and cross-border coordination WHO rapid risk assessment PDF .	Low– Moderate

2.3 GRADE Evidence-to-Recommendation Table

Clinical / Public Health Question	Evidence Summary	Quality of Evidence	Strength of Recommendation	Rationale
Should clinicians treat suspected anthrax before lab confirmation?	Anthrax can progress rapidly to life-threatening disease; WHO 2025 emphasizes early diagnosis and appropriate treatment WHO 2025 anthrax implementation guidance .	Moderate	Strong	Delayed treatment increases risk of severe disease/death; antibiotic harms are generally acceptable relative to benefit.
Should diagnostic specimens be collected before antibiotics?	WHO lists lesion, blood, CSF, and site-specific specimens; CDC recommends collecting specimens before antibiotics if this does not delay care WHO clinician implementation guidance PDF , CDC Clinical Overview .	Moderate	Strong	Improves confirmation and public-health response; must not delay life-saving treatment.
Should localized cutaneous anthrax receive single-agent antibiotics?	WHO outbreak guidance supports ciprofloxacin, doxycycline, penicillin, or amoxicillin for localized cutaneous	Low– Moderate	Strong	Cutaneous anthrax is common and usually antibiotic-responsive; evidence is mainly observational but consistent.

Clinical / Public Health Question	Evidence Summary	Quality of Evidence	Strength of Recommendation	Rationale
	disease WHO rapid risk assessment PDF .			
Should systemic anthrax receive combination antimicrobial therapy?	WHO outbreak guidance recommends combination regimens for inhalation anthrax, sepsis, gastrointestinal anthrax, and meningitis WHO rapid risk assessment PDF .	Low	Strong	Mortality risk is high; combination therapy targets bacteremia, toxin production, CNS penetration, and severe disease.
Should antitoxin be added for severe systemic/inhalation anthrax where available?	CDC 2023 and Mayo Clinic support antitoxin with antibiotics for severe toxin-mediated disease; WHO 2025 does not address specific antitoxins CDC 2023 MMWR , Mayo Clinic .	Low	Conditional	Human efficacy data are limited; biologic plausibility and animal/human supportive data favor use in severe disease where accessible.
Should exposed persons receive PEP?	WHO 2025 includes PEP; CDC supports antibiotics plus vaccine for exposed persons and antibiotics up to 60 days after inhalation exposure risk WHO 2025 anthrax implementation guidance , CDC Clinical Overview .	Moderate	Strong	High consequence of inhalation anthrax and prolonged spore dormancy justify prophylaxis.
Should anthrax response use One	WHO 2008 and WHO outbreak risk assessment	Moderate	Strong	Anthrax is zoonotic and environmental; human control is unlikely

Clinical / Public Health Question	Evidence Summary	Quality of Evidence	Strength of Recommendation	Rationale
Health surveillance and control?	emphasize education, diagnosis, reporting, livestock vaccination, carcass disposal, laboratory confirmation, and human–animal–environment coordination WHO IRIS anthrax PDF , WHO rapid risk assessment PDF .			without animal/environmental source control.

2.4 Updated Recommendations Based on New Evidence

Original WHO-Based Recommendation	New Evidence Impact as of 29 April 2026	Updated Recommendation
Use WHO 2025 clinician guidance for naturally occurring anthrax diagnosis, antibiotics, PEP, and IPC.	No post-26 August 2025 evidence requiring change identified.	Continue to use WHO 2025 guidance as the latest WHO clinical implementation document.
Treat localized cutaneous anthrax with active single-agent antibiotics when uncomplicated.	Mayo Clinic/MSD 2025–2026 updates remain consistent with this approach.	Maintain single-agent therapy for uncomplicated localized cutaneous disease; extend/modify if systemic signs, extensive edema, meningitis concern, or inhalational exposure risk.
Treat systemic anthrax with combination antibiotics and supportive care.	No new contrary evidence identified; CDC 2023 systematic-review-based guidance supports aggressive treatment.	Maintain strong recommendation for immediate inpatient combination therapy for systemic anthrax.
Provide PEP for exposed persons.	CDC guidance continues to support antibiotics plus vaccine where available/indicated and prolonged antibiotics after inhalational exposure.	Maintain PEP recommendation; follow national/public-health authority protocols for antibiotic choice, duration, and vaccine availability.
Use standard IPC precautions and recognize limited person-to-person transmission.	No new evidence indicating change.	Continue standard precautions; handle lesion drainage, contaminated materials, and specimens safely.

Original WHO-Based Recommendation	New Evidence Impact as of 29 April 2026	Updated Recommendation
Implement One Health surveillance and control.	WHO outbreak risk-assessment experience reinforces this approach.	Maintain strong One Health recommendation: human surveillance, veterinary investigation, livestock vaccination, carcass disposal, RCCE, and cross-border notification where applicable.
Consider antitoxin for severe systemic/inhalation anthrax.	WHO 2025 does not specify antitoxins; CDC/Mayo support use where available.	Conditional recommendation: add antitoxin for severe systemic/inhalation disease where accessible and consistent with national protocols.

2.5 Evidence Gaps and Future Research Needs

1. **Formal WHO GRADE tables for anthrax:** WHO anthrax documents located in the summaries do not provide explicit GRADE evidence profiles.
2. **Human comparative treatment trials:** Anthrax rarity makes RCTs difficult; most human evidence remains case reports, case series, and outbreak observations.
3. **Optimal systemic-anthrax combinations:** More evidence is needed on best antimicrobial combinations by syndrome, especially meningitis, gastrointestinal anthrax, and injection-related disease.
4. **Antitoxin effectiveness in humans:** Antitoxin recommendations rely heavily on animal data, biologic plausibility, and limited human observational evidence.
5. **Post-exposure prophylaxis in naturally occurring outbreaks:** More operational evidence is needed on optimal PEP strategies in resource-limited rural settings.
6. **Implementation research:** Studies should assess field uptake of WHO 2025 job aids, diagnostic pathways, antimicrobial availability, and referral systems.
7. **One Health cost-effectiveness:** More evidence is needed on livestock vaccination strategies, carcass-disposal interventions, and community engagement models.
8. **Youth and community engagement in anthrax guidance:** WHO has documented meaningful youth engagement in the abortion care guideline process, but no evidence was found that a similar approach was used for anthrax guidance [WHO case study page](#).

2.6 Final Dynamic Guideline Position as of 29 April 2026

As of **29 April 2026**, the latest WHO anthrax-specific clinical guidance is **WHO's 26 August 2025 Anthrax: implementation guidance for clinicians**. It should be used with the comprehensive 2008 WHO *Anthrax in humans and animals* guideline and WHO outbreak/IDSR materials for surveillance and One Health control.

The practical standard of care is:

1. **Recognize exposure and syndrome early.**
2. **Collect appropriate specimens before antibiotics if this does not delay care.**

3. **Start antibiotics immediately when anthrax is suspected.**
4. **Use single-agent therapy only for uncomplicated localized cutaneous disease.**
5. **Use inpatient combination therapy and supportive care for systemic disease.**
6. **Consider antitoxin for severe systemic/inhalation anthrax where available under national protocols.**
7. **Provide PEP for exposed persons as directed by public-health authorities.**
8. **Notify public health and implement One Health control measures, including animal investigation, livestock vaccination in risk areas, safe carcass disposal, risk communication, and cross-border coordination where needed.**